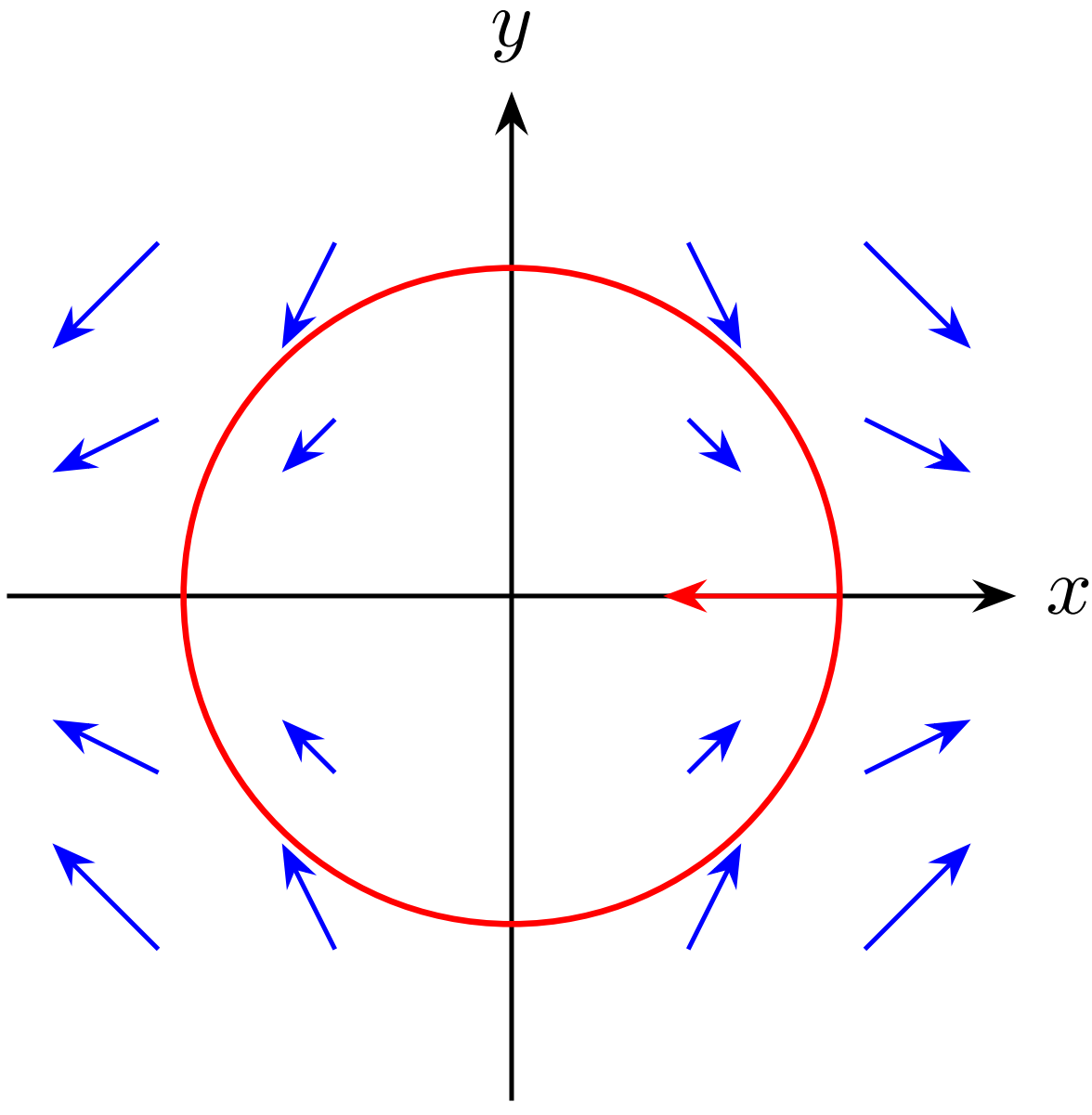

$$\mathbf{B}_\perp = K(x, -y), \quad \mathbf{F}_\perp = -\mu K^2(x, y)/B_0$$